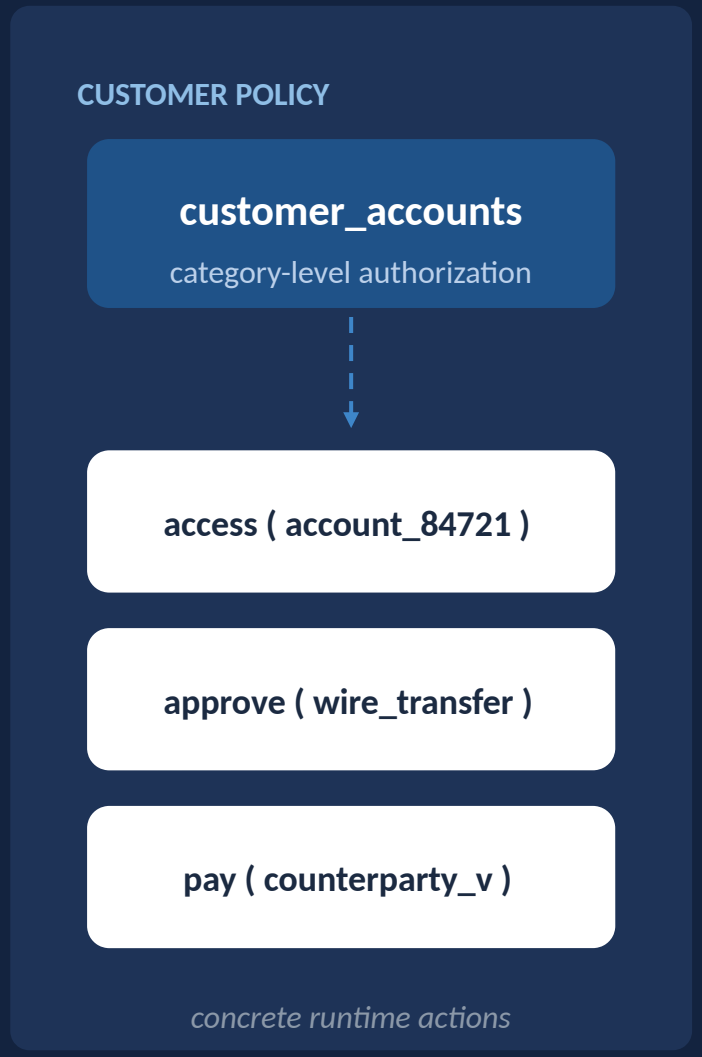
IDV (Intent Drift Verification): Customer-Context-Aware Multi-Engine Architecture for AI-Agent Intent Drift Detection in Finance

The governance problem: broad customer authorization must be applied to concrete runtime actions.

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Computational Intelligence for Financial Engineering & Economics



THE PROBLEM

Policy is written in categories; agents act on instances

INTENT ANCHOR OBJECT · IAO

customer_accounts

Structured, machine-readable representation of approved intent

who owns this mapping?

RUNTIME EVENT

access (account_84721)

A concrete, instance-level action a monitor must judge

Policy defines the category. Runtime monitoring must decide whether a specific instance belongs inside that authorized boundary.

THE PROBLEM

Why finance makes the category→instance problem hard

customer_accounts	which specific account is in scope?
approved_transactions	which amount, to which counterparty?
authorized_trading	which instrument, venue, and size?
approval_authority	does a prior approval still apply?
cumulative_limits	individually valid actions, invalid in aggregate

Finance is hard not because the actions are high-value — but because authorization is contextual, relational, and often cumulative.

THE PROBLEM

Conventional controls answer adjacent questions



Model safety

"Generally harmful?"

Alignment & content filters judge the action in the abstract.



Tool / event guardrail

"Tool call allowed?"

Allowlists and regex filters gate individual API calls.

THE GAP



Customer-relative intent

"Authorized for THIS customer?"

The missing variable: authorized intent vs. observed execution.

IDV complements model safety and tool guardrails by adding customer-specific authorization context.

EMPIRICAL SURPRISE

Two frontier judges disagree on 30% of traces

30%

of traces get opposite drift verdicts

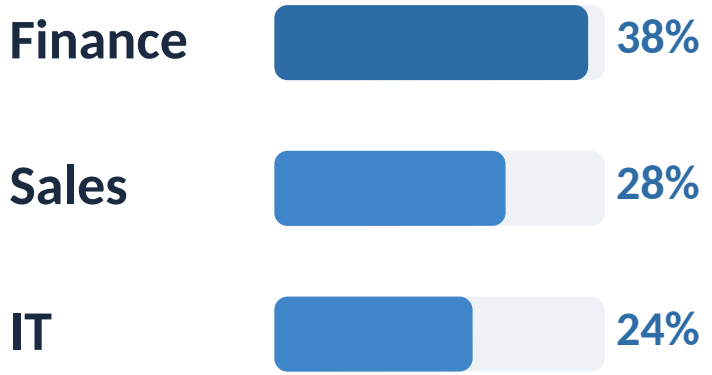
45 of 150 naturalistic traces

38 → 7

one judge flags drift far more often

a strongly asymmetric disagreement pattern

Disagreement by department



Same trace, same customer policy, same task — yet two frontier judges still disagree on where drift begins. That gap is what IDV is built to measure.

Judges: GPT-5.5 and Claude Opus 4.7 · identical traces, IAOs, single-pass judging prompt

EMPIRICAL SURPRISE

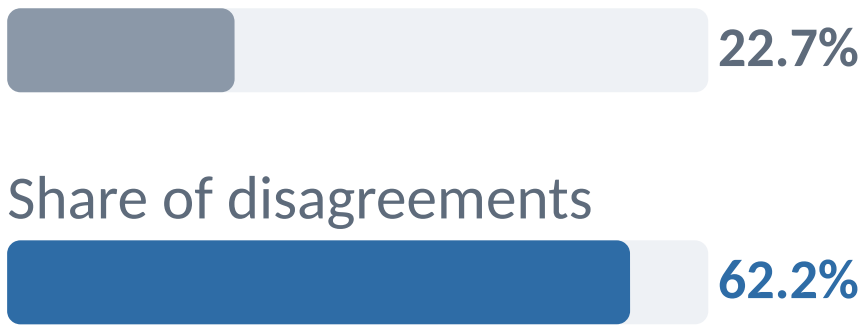
Object-type cases are the locus of divergence

82.4%

P(disagree | GPT-5.5 predicts Object)

vs 30% overall — disagreement is strongly concentrated in Object cases

Object cases are over-represented in disagreements



2.7×

enrichment

of Object cases in the disagreement set

An Object case involves a specific entity under category-level authorization. This boundary region accounts for a disproportionate share of judge disagreement.

STRUCTURAL INTERPRETATION

A single LLM monitor imports a model-specific boundary

SINGLE LLM MONITOR

boundary resolved inside the model

Where the category ends is resolved implicitly inside the model rather than represented as a separately configurable control boundary.

IDV · EXTERNALIZED

- Explicit customer IAO
- Independent detection engines
- Declared thresholds
- Structured audit trail

The claim is not that IDV removes ambiguity — it is that customer-relative interpretation should be explicit and auditable, not silently inherited from one model.

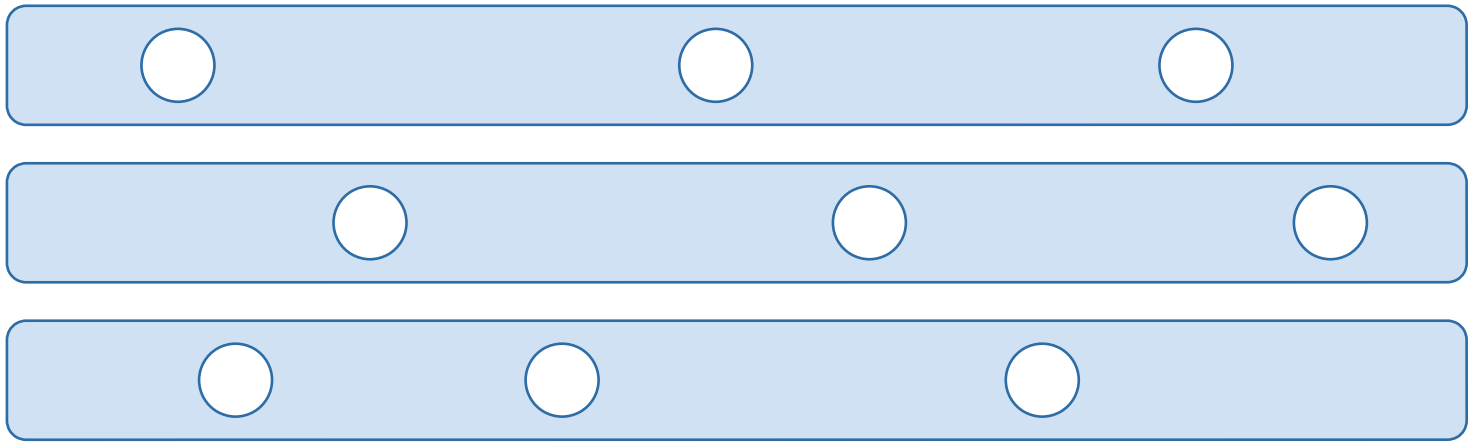
ARCHITECTURE

Design principle: complementary detectors, not a perfect one

Authority
permission / delegation

Coherence
transformer semantic consistency

Scope
entity / resource boundary



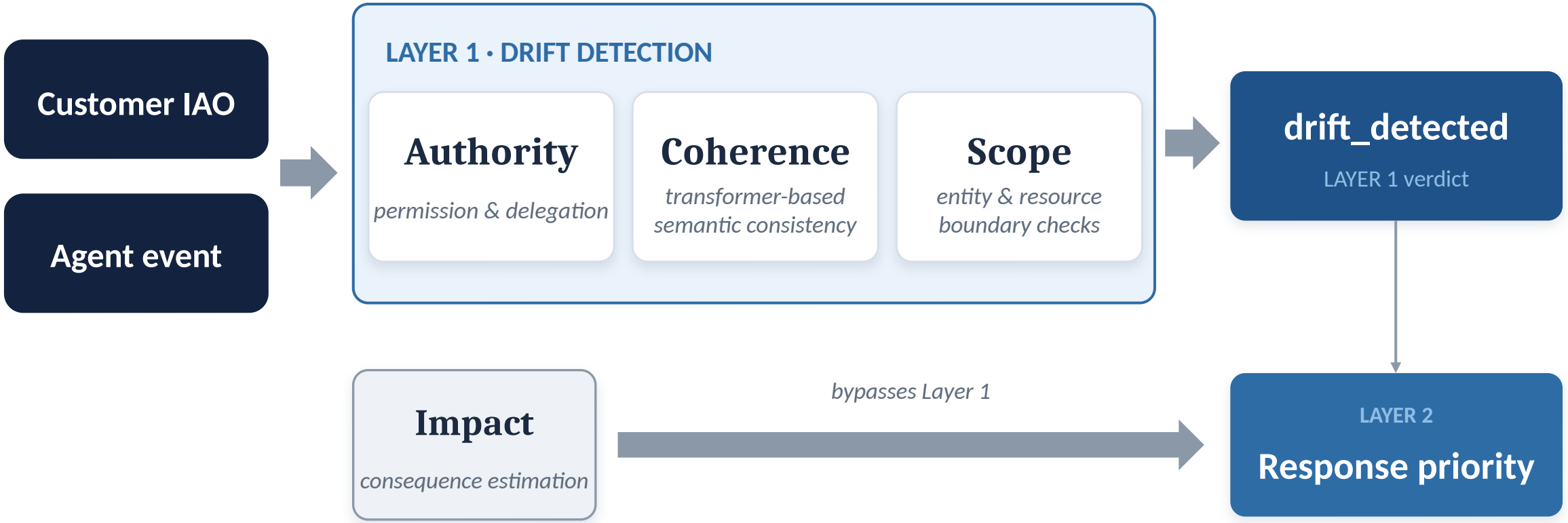
Holes = missed cases. Different detectors fail on different cases.

Impact is separate: it acts after drift detection to prioritize response — it never defines drift.

Defense-in-depth: robustness comes from detectors with different failure modes, not from assuming one perfect detector.

ARCHITECTURE

IDV externalizes intent into a two-layer architecture



The invariant: Impact cannot change drift_detected. Severity ranks the response — it never defines drift.

ARCHITECTURE

High impact is not intent drift



$20M authorized trade

A large, legitimate, in-policy action

HIGH impact **NO drift**

Severity is high, but nothing about the authorization has shifted.



Unauthorized account access

A small action outside the boundary

LOW impact **DRIFT**

Severity is low, but the action crosses the authorized entity boundary.

Impact ranks response severity; it never determines whether drift occurred.

EVALUATION

IDV-1000: controlled performance + a naturalistic probe

850 author-injected controlled traces

- Deterministic perturbation labels
- Three departments: Finance · IT · Sales
- Direct LLM-judge baseline comparison

ROLE · controlled discrimination

150 naturalistic traces

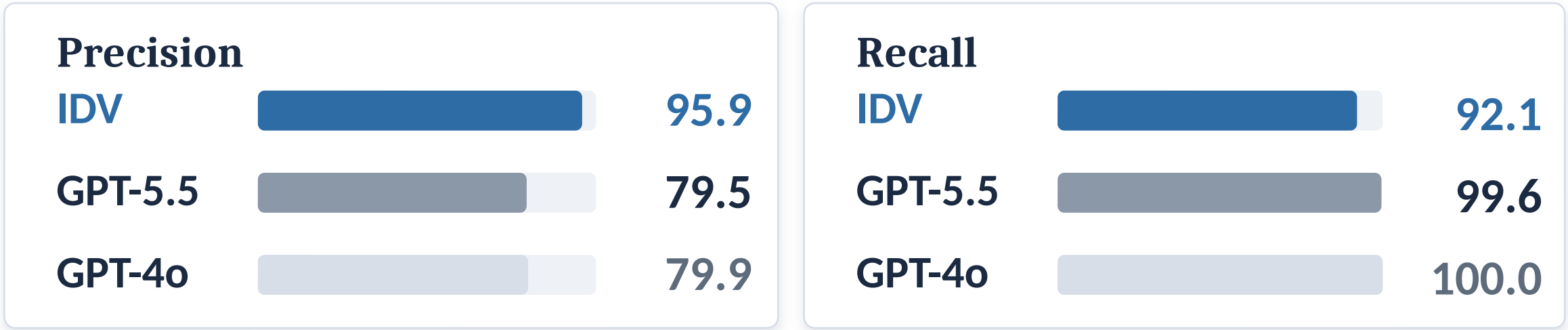
- 30 autonomous pressure scenarios
- Independently generated continuations
- Frontier-judge (LLM) labeling

ROLE · transport & calibration probe

The controlled subset evaluates performance under deterministic labels; the naturalistic subset probes transport, calibration, and judge disagreement.

CONTROLLED PERFORMANCE

IDV occupies a different precision–recall point



The LLM judges are near-perfect-recall detectors. IDV trades a little recall for a large precision gain — and that is what drives alert burden.

CONTROLLED PERFORMANCE

The deployment difference is false-positive burden



6.5×
fewer false positives vs GPT-5.5
(6.4× vs GPT-4o)

All 22 residual IDV false positives occur in Finance, localizing the remaining error burden rather than eliminating it.

CONTROLLED PERFORMANCE

Finance is where the residual ambiguity localizes

38%

frontier-judge disagreement on Finance traces

22 / 22

IDV's residual false positives fall in Finance

4 reverse

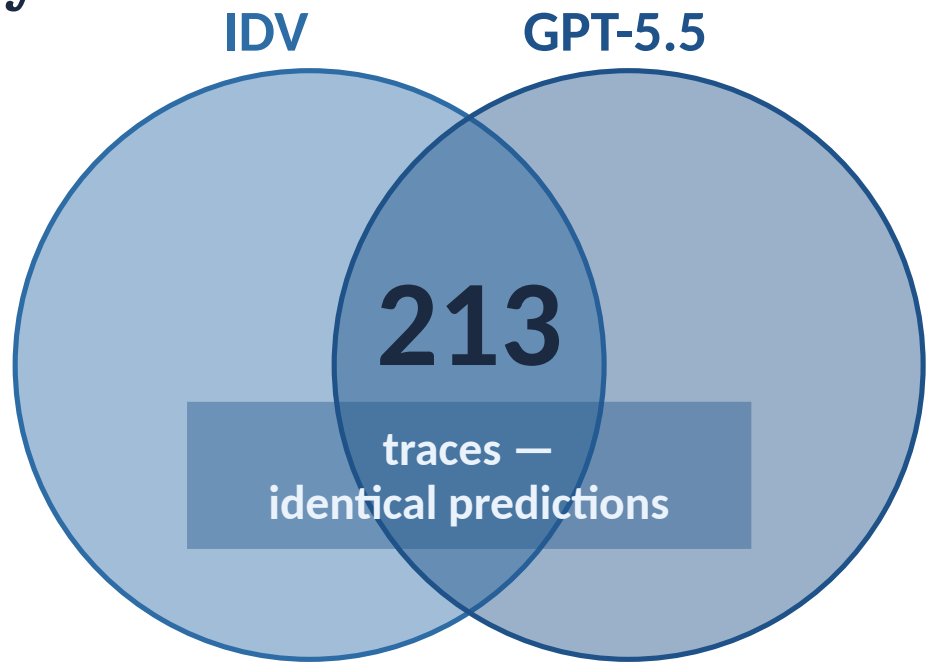
Finance also has the most bidirectional disagreement (IT 1 · Sales 2)

The same general boundary region is difficult for both IDV and the frontier judges. We read this as localization of residual ambiguity, not proof that all Finance traffic is ambiguous.

Bidirectional disagreement also rules out a simple "just use the other judge" remediation in this sample.

CONTROLLED PERFORMANCE

On the 213-case completed-CoT subset, IDV and GPT-5.5 converge exactly



F1 = 96.2%

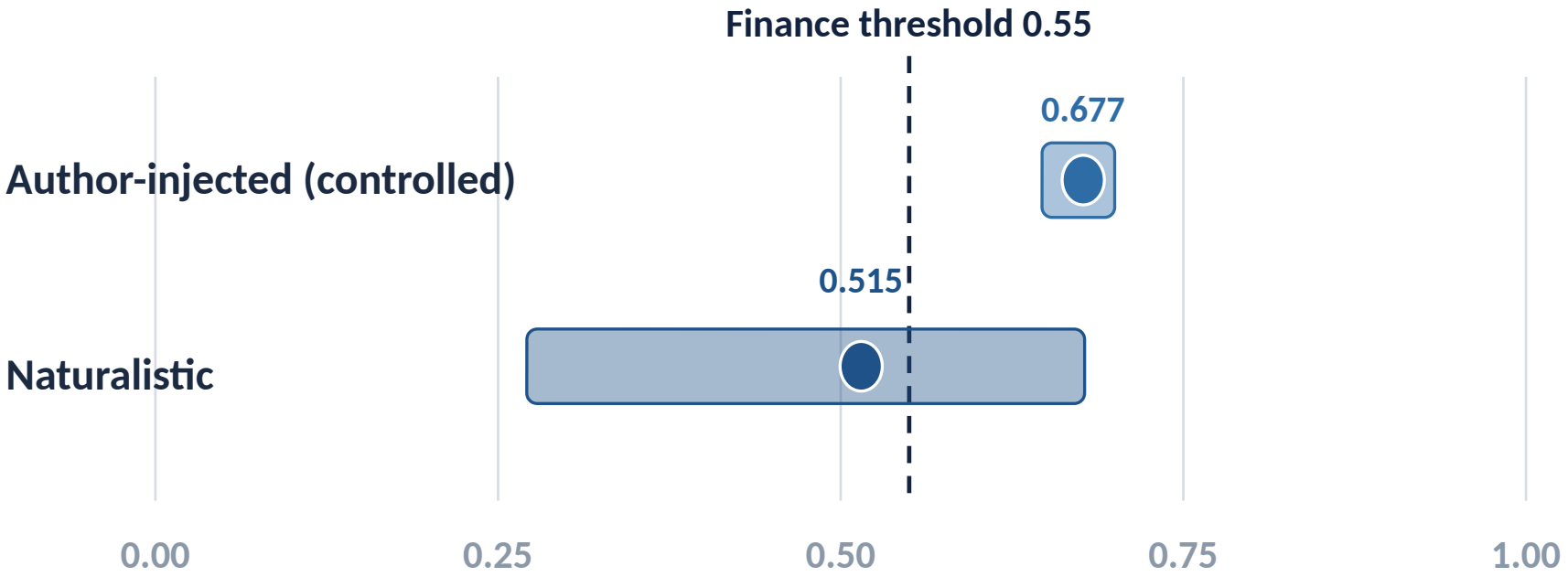
identical for both systems

175	TP	14	FP
24	TN	0	FN

Subset composition: 99% Finance · 64% T1 · zero Object cases. The IDV–LLM gap is concentrated outside this subset, particularly in Object cases and higher-complexity regions.

NATURALISTIC SHIFT

Clean naturalistic events move toward the boundary



AUROC

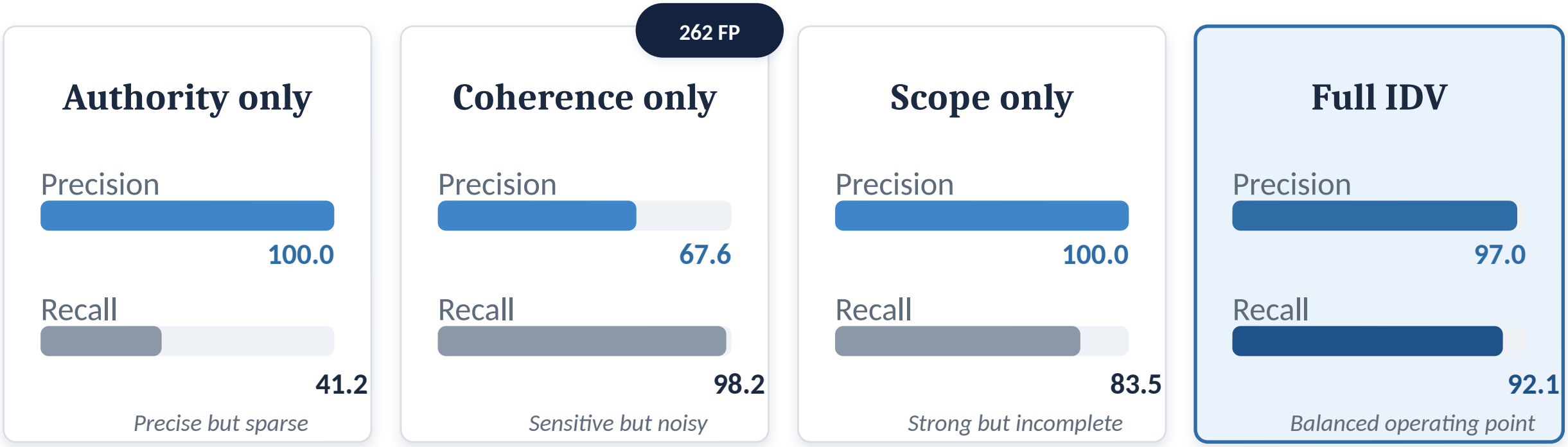
92.3%

Discrimination transfers better than calibration.

Discrimination transfers better than calibration: IDV still ranks cases well (AUROC 92.3%), but the fixed 0.55 Finance threshold does not transfer as the IAO becomes less explicit.

DESIGN VALIDATION

The engines fail differently



Defense-in-depth: independent detectors have different failure modes, so no single weakness determines the final decision.

DESIGN VALIDATION

Hybrid trades controlled F1 for better naturalistic ranking

Weighted-mean adjudicator

zero-FP option for lower-risk profiles

Controlled F1	95.7%
Precision	100.0%
False positives	0
Naturalistic AUROC	72.4%

Full IDV — hybrid blend

better naturalistic ranking

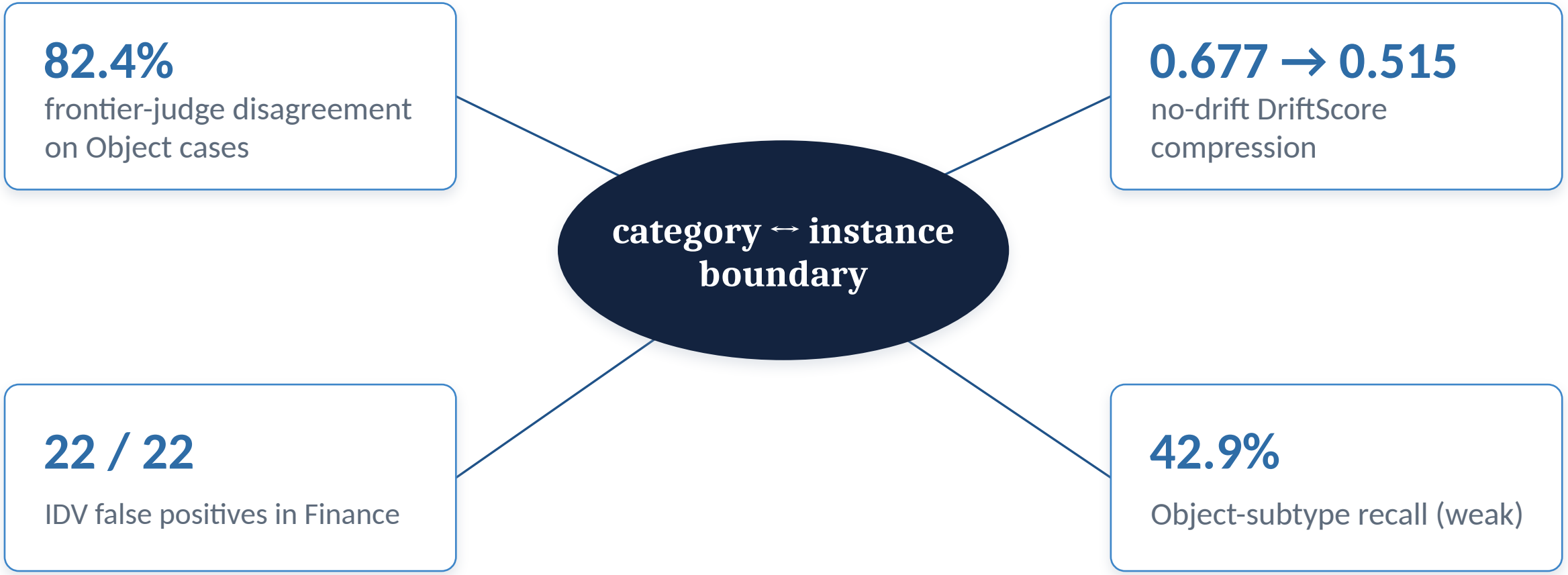
Controlled F1	94.5%
Precision	97.0%
False positives	16
Naturalistic AUROC	81.8%

On the controlled ablation, the weighted mean wins at the fixed threshold. The hybrid gives better ranking on the small naturalistic ablation, which matters because Layer 2 uses a continuous score for prioritization.

Ablation on N=818 controlled · naturalistic AUROC on N=100 (9 drift) — directional only, not comparable to the 92.3% headline.

SYNTHESIS

Four results converge on the same boundary region



Across distinct analyses, difficulty repeatedly concentrates where category-level authorization must be applied to instance-level actions.

GOVERNANCE IMPLICATION

Who owns the authorization boundary?

1. **Category → instance** is the core measurement problem — where authorization is actually resolved.

2. **Single-LLM monitoring leaves the authorization boundary implicit and model-dependent.**

3. **An external architecture makes the boundary explicit: higher precision, structured audit trails, and per-profile calibration.**

DETECT WHEN IT SHOULD NOT DECIDE



Externalize
the boundary



Calibrate
per customer profile



Refine
the IAO over time



Escalate
genuine ambiguity

The conclusion is not "build a stronger classifier." It is to make the authorization boundary observable — and then govern it.

Q & A

Thank You!